



ORIGINAL ARTICLE

Comparison of effect of kangaroo mother care, breastfeeding and swaddling on Bacillus Calmette-Guerin vaccination pain score in healthy term neonates by a clinical trial

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Abstract

Background/aim: The purpose of this research was to compare the analgesic effect of kangaroo mother care (KMC), breastfeeding and swaddling in Bacillus Calmette-Guerin (BCG) vaccination in term neonates.

Methods: In a randomized 120 healthy term neonates who received routine BCG vaccination in the first day of their life are distributed into three groups. In group 1, neonates breastfed two minutes before, during and one minute after BCG vaccination. In group 2, neonates received KMC 10 minutes before, during and one minute after vaccination and in group 3, they were swaddled 10 minutes before, during and one minute after vaccination. Primary outcomes included pain score during, one minute and two minutes after BCG vaccination and obtaining pain score of less than three during vaccination.

Results: Pain scores during, one minute and two minutes after vaccination in group 1 were lower than in groups 2 and 3. Group 1 had higher success rate in painless vaccination and had lower crying duration in comparison to another groups ($p < 0.05$).

Conclusion: Breastfeeding was more effective than KMC and swaddling in reduction of BCG vaccination pain in healthy term neonates.

Keywords

Breastfeeding, kangaroo mother care, neonate, swaddling, vaccination pain

History

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Introduction

Neonates can experience pain and they react to pain by autonomic changes such as increase in heart rate and blood pressure, alternation in cortisol and catecholamine hormones and behavioral responses [1]. Harming complications of frequent painful exposures in neonates include change in pain sensitivity, which may continue up to adolescence, constant neuroanatomical and behavioral abnormalities and long-term adverse effects such as emotional, behavioral and learning disabilities [2].

In neonates, nonpharmacologic interventions such as nonnutritive sucking with or without sucrose or glucose, breastfeeding, swaddling, kangaroo mother care (KMC) and sensorial saturation (use of touch, massage, voice, smell and sight) has been shown to reduce mild to moderate pain of minor procedures in both preterm and term newborns [3,4].

Close skin-to-skin contact between mother and infant in KMC method and resting of neonate between the mother's breasts can reduce stress and pain responses of neonates by

facilitating maternal-infant co-regulation and stimulating of ventral tactile and proprioceptive systems and it is a simple, effective and safe pain relieving method for single painful procedure [5,6].

Swaddling by different techniques in many countries can alleviate pain, improve neuromuscular development and reduce physiologic distress in preterm infants and also might decrease excessive crying in children with cerebral damage [7].

In term neonates, breastfeeding or oral formula feeding is more effective than oral sucrose or glucose for pain control of heel-lancing [8].

In healthy preterm neonates, KMC is more effective than 25% glucose to reduce pain of heel lancing [9].

The routine vaccines that may be injected intramuscularly, subcutaneously or intradermal are the most common source of iatrogenic pain during infancy and childhood [10]. Control, management and decreasing of pain of immunization and other minor procedures of routine medical and nursing care of neonates are necessary [2].

Theory/calculation

According to vaccination programs of Ministry of Health and Medical Education of I.R. Iran, all neonates should receive

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intradermal injection of Bacillus Calmette-Guerin (BCG) vaccine in hospitals which they are delivered, routinely in the first day of their life. Since other studies have proven that breastfeeding, swaddling and kangaroo mother care are more effective than sucrose or glucose in reducing of pain [5–9], therefore, this research was done to answer this question that which one of these three non-pharmacological interventions (KMC, breastfeeding and swaddling), can cause less pain of intradermal BCG vaccination and the purpose of this study was to compare analgesic effect of KMC, breastfeeding and swaddling in BCG vaccination of healthy term neonates.

Materials and methods

In a randomized single-blind clinical, parallel group trial, all healthy term neonates who received routine BCG vaccination in the first day of their life in Shahid Sadoughi Hospital, Yazd, Iran from March to June 2015 were enrolled in the study.

Sample size was assessed to be 37 neonates in each group based on Z formula and a confidence interval of 95% with 80% power to detect a difference of 30% in reduced pain vaccination between the groups with type one error (alpha) of 0.05 and 40 neonates in each group were recruited to account for missing data and dropouts.

Eligible participants included term neonates (gestation age of 37–42 weeks) who were product of normal vaginal delivery, awake and alert before vaccination, without systemic illness, in a healthy medical condition and with birth weight of 2500–4000 g. Exclusion criteria consisted of moderate to severe birth asphyxia, severe congenital malformation and receiving sedative hypnotic or analgesic drugs within the past 24 h.

The trial used computer generated equal simple randomization by random numbers and allocation ratio was 1:1 for the three groups.

Since, neonates usually breastfeed within 3–5 minutes and length of swaddling and KMC should be at least 10 minutes, and interventions and their length were different, so, blinding of parents of the participants, the nurse of hospital, the Data collector and the outcome assessor, was not possible and data analysts were only kept blinded to the allocation. But, concealment was done by writing down the intervention for each serially participating neonate in a numbered and sealed opaque envelope which was opened by the pediatric neurologist of research immediately before intervention. Randomization and concealment were done by a researcher with no clinical involvement in the trial.

The neonates were randomly distributed into three groups. In group 1, breastfeeding started two minutes before vaccination and continued during and one minute after BCG vaccination. In group 2, neonates received KMC 10 minutes before, during and one minute after BCG vaccination and in group 3, they were swaddled 10 minutes before, during and one minute after vaccination.

The neonates were awake, quiet and alert and had clean diapers at the time of vaccination.

The BCG vaccine of the research were products of Pasteur Institute of Iran and in all of the neonates, the vaccine was injected in similar conditions, similar needles and by a trained nurse. Dosage of the BCG vaccine was 0.5 ml that was

injected intradermal with a 27 mm gauge needle into the area of right deltoid muscle, at a 90° angle.

In KMC group, the mother hold the neonate wearing only a diaper under her gown between her bare breasts in an upright position with maximum skin-to-skin, i.e. chest-to-chest contact.

Hip and knees of the neonates in swaddling group, were wrapped in the blanket with mild tightness and their hands and heads were free.

The intervention was delivered by a nurse of hospital and primary and secondary outcomes were assessed by the pediatric resident of research.

Primary outcomes included baseline pain score five seconds before, during, one minute and two minutes after BCG B vaccination, and obtaining of pain score of less than three during BCG vaccination (less pain vaccination). Pain score when the needle was inserted to the skin, was assessed based on Neonatal/Infant Pain Scale (NIPS). Parameters of NIPS include facial expression, cry, breathing patterns, position of arms, position of legs and state of arousal. The score for each item is summed and total score ranges from 0 to 7. It is recommended for children less than one year of age and a score greater than three indicates pain [11].

Obtaining pain score less than three based on NIPS during the needle insertion to skin was considered successful in reducing the pain and less pain procedure.

Secondary outcome was duration of neonate crying during BCG vaccination that was calculated from the onset of crying (when the needle was inserted) until crying was stopped for five seconds.

The data were analyzed using Statistical Package for the Social Sciences version 17 (SPSS, Chicago, IL) statistical software. Recorded data were assessed for normal distribution using the Kolmogorov–Smirnov test and Chi-square test was used for data analysis of categorical variables and continuous and mean variables were compared using one-way analysis of variance (ANOVA) test between the three groups. Differences were considered significant at *p* values of less than 0.05.

Informed consent was taken from the parents of the participating neonates before enrolling and this research has been approved by the Ethics Committee of Shahid Sadoughi University of Medical Sciences, Yazd, Iran.

This research is registered in Iranian clinical trials with registration number: IRCT201505142639N16.

Results

The design and conduct of this trial was straightforward, we did not have any losses or exclusions and 50 girls and 70 boys with mean birth weight of 3.14 ± 0.33 kg were evaluated in the three groups.

By Kolmogorov–Smirnov test, the data had normal distribution.

Comparison of demographic characteristics of the children in the three groups is shown in Table 1, which indicates that no statistically significant differences were seen from viewpoints of sex distribution, mean of weight and means of height.

Table 2 shows comparison of pain scores in the three groups and indicates that baseline pain score before BCG

Table 1. Comparison of demographic characteristics of children in both groups.

Groups Data	Breastfeeding	Kangaroo mother care	Swaddling	<i>p</i> value
Sex				
Girl	16	17	17	0.8
Boy	24	23	23	
Weight in kg (mean $\pm$ SD)	3.16 $\pm$ 0.32	3.15 $\pm$ 0.34	3.11 $\pm$ 0.31	0.7
Height in centimeter (mean $\pm$ SD)	50.22 $\pm$ 1.77	50.01 $\pm$ 1.51	49.38 $\pm$ 2.44	0.6

Table 2. Comparison of pain scores in the three groups.

Groups Data	Breastfeeding	Kangaroo mother care	Swaddling	<i>p</i> value
Pain score before the vaccination (mean $\pm$ SD)	0	0	0	1
Pain score during BCG vaccination (mean $\pm$ SD)	1.81 $\pm$ 1.16	2.92 $\pm$ 1.34	3.48 $\pm$ 0.96	0.001
Pain score 1 min after vaccination (mean $\pm$ SD)	0.79 $\pm$ 0.3	1.19 $\pm$ 0.97	1.61 $\pm$ 1.01	0.001
Pain score 2 min after vaccination (mean $\pm$ SD)	0.31 $\pm$ 0.1	0.73 $\pm$ 0.36	1.07 $\pm$ 0.56	0.04

Table 3. Comparison of frequency of doing less pain BCG vaccination and BCG injection crying duration in three groups.

Groups Data	Breastfeeding	Kangaroo mother care	Swaddling	<i>p</i> value
Less pain BCG vaccination				
Yes	40	38	36	0.04
No	0	2	4	
Duration of baby crying in seconds (mean $\pm$ SD)	26.61 $\pm$ 11.7	45.12 $\pm$ 23.41	63.25 $\pm$ 16.98	0.001

vaccination was not statistically significant different in three groups, but pain scores of during, one minute and two minutes after BCG vaccination in breastfeeding group were lower than in kangaroo mother care or swaddling groups.

Comparison of frequency of doing less pain BCG vaccination (obtaining of pain score of less than three during BCG vaccination) and duration of baby crying during vaccination are presented in Table 3, which shows that breastfeeding group had higher success rate in reduced pain vaccination and lower crying duration in comparison to kangaroo mother care or swaddling groups.

Discussion

Non-pharmacological methods are the first choice for analgesia in newborns immunization [12] and this research showed that breastfeeding was more efficient than KMC and swaddling in reduction of BCG vaccination pain. Cignacco et al. in their systematic review of 13 randomized controlled studies and two meta-analyses, concluded that only non-nutritive sucking, swaddling and facilitated tucking had reducing pain effect [13].

In a study in Tabriz, Iran, breastfeeding was more efficient than oral sucrose, combination of oral sucrose and breastfeeding and no intervention in reduction of pain of intramuscular diphtheria, tetanus, pertussis (DTP) vaccination in less than three months old infants [14].

In a study in Mashhad, Iran, breastfeeding was more efficient than, kangaroo mother care or no intervention in reduction of pain of intramuscular DTP vaccination in two, four and six months old infants [15].

In a study in Tehran, Iran, KCM 10 minutes before and during intramuscular injection in healthy term neonates was more effective than no intervention in reduction of cumulative Neonatal Infant Pain Scale score immediately after injection [16].

Kostandy et al. study showed analgesic effect of skin-to-skin contact in hepatitis B vaccination in healthy term neonates [17]. A study in Quebec, Canada, analgesic effect of KMC for 30 min before routine heel sticking of preterm neonates within 10 days of birth, was more than swaddling [18].

Possible explanation for the discrepancy may be related to differences in age, procedures, type of vaccine and route of injection.

In present research, breastfeeding was the most efficient analgesia in neonatal BCG vaccination and analgesic effects of breastfeeding in reducing of vaccination pain in less than six months old infants [19], in healthy term neonates during routine heel sticking [20] and venipuncture [21,22] have been shown in other researches.

Analgesic and lowering stress effect of breastfeeding might be due to its helping mothers and neonates to cope with the stressful situations. On the other hand, breastfeeding can amend vaccines response, decrease risk of unsuitable reactions of vaccines in sensitive infants and change metabolism of ethyl mercury derived from some vaccines [23].

Some research showed that non-pharmacological pain management methods in newborns were much more efficient when used in combination with other non-pharmacological interventions including music therapy, swaddling, facilitated tucking, KMC and non-nutritive suction [24].

In conclusion, based on the result of the present study, breastfeeding was more effective than KMC and swaddling in reduction of BCG vaccination pain in healthy term neonates, but, skin-to-skin contact or KMC, also decreased immunization pain and therefore, combination of breastfeeding and kangaroo mother care may be more effective. So, health provider can alleviate vaccination pain in neonates with this simple, easy, cost effective and available strategy.

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Declaration of interest

No conflicts of interest declared. This study was funded by a grant from the Deputy for Research of Shahid Sadoughi University of Medical Sciences, Yazd, Iran.

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